Semantic Similarity Index

Measuring the similarity of semantic fields with ARI

Store a dataset of circa 1.8M word associations (e.g. "ship"→"sail") in hetero-associative memory. Retrieve the activation map for multiple words. Measure their similarity via the Adjusted Rand Index.

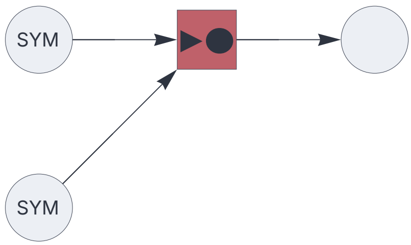
Requirements

```
<< "Circuits.m"
```

Circuit

```
{
  "$schema": "https://creatingintelligence.org/schemas/circuit-v1.json",

  "hyperparameters": {"default": [5000, 10]},

  "dataflow": [
    {"component": "input", "plugin": "codec", "send": [1]},

    {"component": "input", "plugin": "codec", "send": [2]},

    {"component": "associator", "send": [11], "receive": [2, 1]},

    {"component": "output", "receive": [11]}
  ]
}
```

```
circ = Circuit[json];
f = circ["function"];
circ["schematics"][ImageSize→220]
```



Import dataset

```
data = Import[ NotebookDirectory[] <>  "/data/word-associations.csv.zip",
         "word-associations.csv"];
```

```
RandomSample[data, 10]
```

```
{{centralism, system}, {solutions, replacement},
 {appraised, performance}, {dispersant, clean-up},
 {lede, beginning}, {waterlogged, soil}, {antimatter, symmetry},
 {squalidness, wretchedness}, {hundred, cent}, {lapses, missteps}}
```

```
Length[data]
```

```
1 888 505
```

Train

```
f @@@ data;
```

Adjusted Rand Index

```
ARI[A_List, B_List, dim_Integer] :=
    N[(Length[Intersection[A,B]] - Length[A]*Length[B] / dim ) /
            (Min[Length[A], Length[B]] -  Length[A]*Length[B] / dim)]
```

```
similarity[x_, y_] := ARI[ f[x, Null], f[y, Null], 5000]
```

Test

Retrieve activation map.

```
f["cat", Null]
```

```
Sequence[{9, 12, 25, 29, 41, 49, 65, 68, 69, 70, 72, 90, 124, 126, 131, 141,
  154, 165, 174, 179, 181, 183, 194, 209, 219, 223, 226, 233, 237, 249,
  257, 272, 277, 286, 287, 300, 301, 318, 334, 341, 346, 362, 368, 374,
  381, 387, 410, 416, 431, 451, 452, 457, 458, 462, 463, 466, 476, 479,
  507, 509, 517, 518, 519, 534, 538, 539, 540, 544, 552, 563, 570, 603,
  604, 610, 618, 630, 645, 649, 664, 668, 674, 687, 702, 710, 730, 743,
  756, 758, 768, 793, 800, 833, 835, 841, 855, 863, 881, 897, 898, 901,
  902, 904, 906, 909, 916, 923, 932, 934, 946, 948, 966, 975, 981, 983,
  985, 991, 995, 1000, 1003, 1008, 1019, 1033, 1040, 1043, 1045, 1054,
  1055, 1066, 1068, 1109, 1118, 1131, 1132, 1134, 1166, 1168, 1169, 1174,
  1185, 1186, 1188, 1206, 1222, 1249, 1286, 1289, 1294, 1350, 1354, 1355,
  1356, 1361, 1363, 1377, 1387, 1389, 1412, 1419, 1438, 1481, 1482, 1504,
  1516, 1526, 1533, 1537, 1547, 1553, 1568, 1592, 1601, 1625, 1626, 1634,
  1637, 1641, 1642, 1643, 1666, 1683, 1700, 1716, 1758, 1773, 1796, 1812,
  1821, 1824, 1826, 1830, 1842, 1847, 1861, 1863, 1869, 1874, 1887, 1890,
  1891, 1892, 1911, 1926, 1929, 1945, 1949, 1985, 2019, 2034, 2088, 2089,
  2091, 2094, 2124, 2135, 2142, 2144, 2151, 2167, 2172, 2177, 2195, 2200,
  2210, 2211, 2214, 2215, 2238, 2242, 2244, 2247, 2264, 2290, 2292, 2308,
  2329, 2348, 2354, 2367, 2370, 2380, 2394, 2410, 2419, 2427, 2428, 2431,
  2432, 2434, 2450, 2470, 2473, 2489, 2492, 2512, 2532, 2533, 2547,
  2550, 2565, 2576, 2607, 2612, 2617, 2637, 2641, 2651, 2660, 2663,
  2666, 2668, 2670, 2682, 2684, 2686, 2688, 2692, 2710, 2715, 2721, 2740,
  2746, 2750, 2751, 2754, 2764, 2769, 2770, 2788, 2793, 2797, 2801, 2814,
  2824, 2831, 2851, 2892, 2901, 2904, 2908, 2921, 2928, 2929, 2930, 2931,
  2932, 2937, 2942, 2954, 2977, 2999, 3000, 3016, 3039, 3042, 3053, 3064,
  3074, 3077, 3084, 3092, 3097, 3121, 3124, 3129, 3130, 3136, 3137, 3147,
  3152, 3154, 3155, 3160, 3177, 3196, 3200, 3211, 3235, 3257, 3258, 3264,
  3271, 3280, 3289, 3302, 3310, 3311, 3318, 3321, 3332, 3337, 3354, 3356,
  3361, 3373, 3374, 3380, 3383, 3388, 3398, 3411, 3453, 3473, 3503, 3504,
  3513, 3530, 3532, 3537, 3549, 3577, 3579, 3593, 3612, 3636, 3637, 3644,
  3652, 3662, 3664, 3682, 3694, 3698, 3708, 3713, 3714, 3718, 3720, 3736,
  3756, 3760, 3761, 3763, 3774, 3813, 3837, 3838, 3843, 3854, 3864, 3870,
```

```
3873, 3875, 3883, 3902, 3903, 3915, 3923, 3938, 3943, 3965, 3972,
3982, 3985, 3994, 4044, 4050, 4070, 4072, 4076, 4086, 4104, 4106,
4123, 4143, 4159, 4165, 4170, 4190, 4197, 4200, 4220, 4232, 4234,
4244, 4266, 4271, 4279, 4288, 4298, 4301, 4308, 4315, 4322, 4323,
4326, 4341, 4347, 4357, 4358, 4366, 4368, 4379, 4394, 4401, 4403,
4407, 4408, 4412, 4418, 4427, 4455, 4459, 4476, 4510, 4511, 4515,
4519, 4538, 4541, 4548, 4557, 4560, 4571, 4574, 4583, 4594, 4603,
4609, 4610, 4618, 4626, 4635, 4648, 4651, 4663, 4680, 4686, 4699,
4709, 4714, 4722, 4726, 4740, 4742, 4744, 4747, 4757, 4759, 4768,
4779, 4787, 4819, 4823, 4830, 4844, 4845, 4866, 4884, 4892, 4894,
4896, 4907, 4919, 4927, 4942, 4949, 4967, 4974, 4986, 4996, 4998}]
```

Measure semantic similarity via ARI:

```
similarity["ship", "boat"]
```

```
0.633625
```

```
similarity["ship", "cheese"]
```

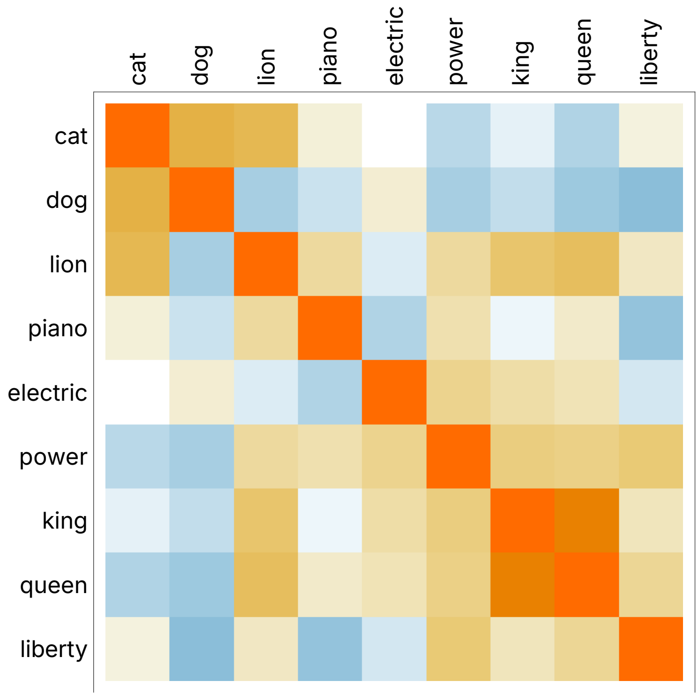
```
-0.00990518
```

Similarity Matrix

```
SimilarityMatrix[ items_List] := Module[ {simtable, xTicks, yTicks},

    simtable = Outer[ Round[similarity[##], 0.001]&, items, items];
    xTicks=Table[{i,Rotate[items⟦i⟧,90 Degree], {0,0}},{i,Length[items]}];
    yTicks=Table[{i,items⟦i⟧, {0,0}},{i,Length[items]}];

    MatrixPlot[simtable,
        FrameTicks → {{yTicks, None}, {None, xTicks}},
         FrameStyle → Opacity[0],
        FrameTicksStyle → Directive[Opacity[1],
        FontFamily → "Inter Variable",
        FontSize→12]]
    ]
```

```
items = {"cat",  "dog", "lion", "piano",
     "electric", "power", "king", "queen", "liberty"};
```

```
SimilarityMatrix[ items]
```




```
(* Export["similaritymatrix.png", ImagePad[Rasterize[%], 30, White]]; *)
```